

Disproof of Conjecture 00000001072: The Prime-Factor Spectrum Claims for the Gaussian-Binomial Counting Function

TLMC submission

September 2026

1 The conjecture

“Let $G(n, k, q)$ denote the number of k -dimensional planes of $\mathbb{F}_q^n$. Conjecture: The prime-factor spectrum of the Gaussian-binomial formula $G(n, k, q) = q^{(n-k)k} \cdot \binom{n}{k}_q$ consists exactly of the primes $\leq q^n - 1$; every prime factor divides some $q^i - 1$ with $i \leq n$, and conversely each such prime occurs for suitable (n, k) .”

We refute the conjecture with two independent counterexamples. Verdict: **FALSE**.

2 Counterexample I: the “every prime factor divides some $q^i - 1$, $i \leq n$ ” claim fails

Take $(n, k, q) = (2, 1, 2)$. Then

$$G(2, 1, 2) = 2^{(2-1) \cdot 1} \cdot \frac{2^2 - 1}{2 - 1} = 2 \cdot 3 = 6,$$

so the prime-factor spectrum of $G(2, 1, 2)$ is $\{2, 3\}$. But

$$q^1 - 1 = 2^1 - 1 = 1, \quad q^2 - 1 = 2^2 - 1 = 3.$$

The prime 2 divides neither 1 nor 3; hence 2 divides no number of the form $q^i - 1$ with $i \leq n = 2$. Nevertheless $2 \mid 6 = G(2, 1, 2)$. The asserted inclusion of the spectrum in the prime divisors of $\{q^i - 1 : i \leq n\}$ therefore fails.

3 Counterexample II: the “exactly the primes $\leq q^n - 1$ ” claim fails

Take $(n, k, q) = (2, 1, 3)$. Then

$$G(2, 1, 3) = 3^{(2-1) \cdot 1} \cdot \frac{3^2 - 1}{3 - 1} = 3 \cdot 4 = 12,$$

whose prime-factor spectrum is $\{2, 3\}$. The primes $\leq q^n - 1 = 3^2 - 1 = 8$ are

$$\{2, 3, 5, 7\}.$$

Both $5 \leq 8$ and $7 \leq 8$, yet $5 \nmid 12$ and $7 \nmid 12$: the primes 5 and 7 do not occur in the spectrum. Hence the spectrum is a *proper subset* of $\{p \text{ prime} : p \leq q^n - 1\}$, and the word “exactly” fails.

Supporting evidence

For $(n, k, q) = (3, 1, 2)$:

$$G(3, 1, 2) = 2^{2 \cdot 1} \cdot \frac{2^3 - 1}{2 - 1} = 4 \cdot 7 = 28,$$

with spectrum $\{2, 7\}$, while the primes $\leq 2^3 - 1 = 7$ are $\{2, 3, 5, 7\}$: the primes 3 and 5 are missing.

4 Summary table

(n, k, q)	$G(n, k, q)$	spectrum	claimed set	violation
$(2, 1, 2)$	$6 = 2 \cdot 3$	$\{2, 3\}$	$\{\text{primes dividing some } q^i - 1, i \leq 2\} = \{3\}$	2 divides no $q^i - 1$
$(2, 1, 3)$	$12 = 2^2 \cdot 3$	$\{2, 3\}$	$\{p \leq 3^2 - 1\} = \{2, 3, 5, 7\}$	5, 7 missing
$(3, 1, 2)$	$28 = 2^2 \cdot 7$	$\{2, 7\}$	$\{p \leq 2^3 - 1\} = \{2, 3, 5, 7\}$	3, 5 missing (supporting)

5 Verification

All arithmetic was independently recomputed. For $k = 1$ the values 6, 12, 28 were additionally confirmed by direct brute-force enumeration of the affine lines of $\mathbb{F}_q^n$ as sets of points, so the counterexamples are robust to the interpretation of “ k -dimensional planes”. A self-contained script (`reproduce.py`) and a zero-axiom Lean 4 certificate (`lean4/`, core only, no Mathlib; all goals closed by `rfl/decide`) accompany this document.

6 Scope and boundaries

We refute only the two directional claims:

1. “every prime factor divides some $q^i - 1$ with $i \leq n$ ” — false at $(2, 1, 2)$;
2. “the spectrum consists exactly of the primes $\leq q^n - 1$ ” — false at $(2, 1, 3)$.

The Gaussian-binomial counting formula $G(n, k, q) = q^{(n-k)k} \cdot \binom{n}{k}_q$ itself is of course correct; it is the conjecture’s own definition of G and is used only as such. The third (“conversely”) clause is not adjudicated here.